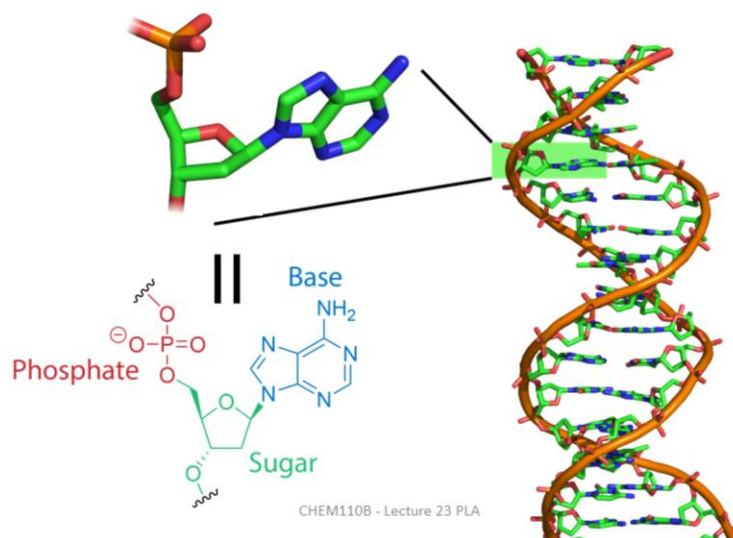
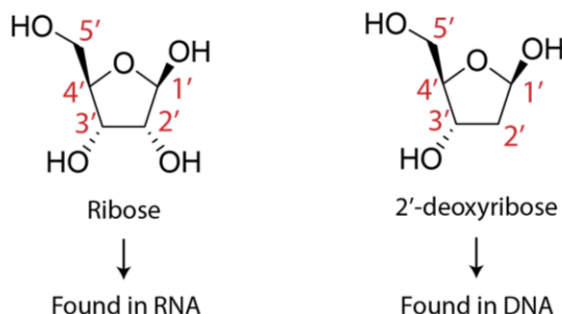


Nucleotides consist of a sugar, a phosphate group, and a nitrogenous base



Nucleic acid is a general term that includes both DNA (deoxyribonucleic acid) and RNA (ribonucleic acid). Nucleic acids get their name because they were originally found in the nucleus, although we now know they are found in other regions of the cell. They called "acids" because their backbone has a phosphoric acid unit. Both DNA and RNA are polymers made of monomers known as nucleotides. Shown here is a DNA double helix, and a single nucleotide is highlighted. The nucleotide structure contains three pieces, a sugar (green in the skeletal structure), a phosphate group (red in the skeletal structure), and a nitrogenous base (blue in the skeletal structure). The nitrogenous base is also called a nucleobase (the two terms mean the same thing). The identity of the nitrogenous base varies from one monomer to the next, and it is the order of monomers that allows nucleic acids to store information. The sugar and phosphate, which comprise the so-called "sugar-phosphate" backbone, are invariant, with the exception that the identity of the sugar determines whether the nucleic acid is RNA or DNA. Another defining difference between DNA and RNA is that DNA in the cell is always double-stranded, while RNA is single-stranded.

The structure of the sugar determines whether a nucleotide chain is DNA or RNA



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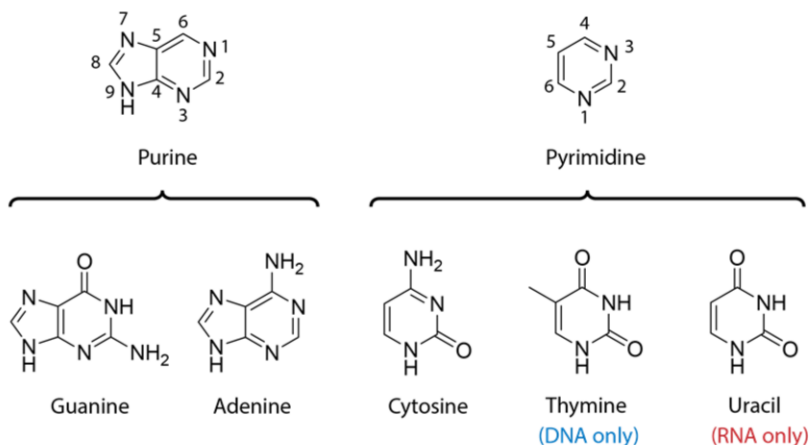
2

The two different sugars that are found in polynucleotide chains (nucleic acids) are ribose and 2'-deoxyribose (or deoxyribose for short). The two sugars are chemically identical except that the 2' carbon in deoxyribose is not bonded to a hydroxyl (i.e., OH) group. In other words, the 2' -OH is changed to a 2' hydrogen atom giving rise to the prefix "deoxy" for the sugar in DNA. Ribose is the sugar found in RNA nucleotides and 2'-deoxyribose is the sugar found in DNA nucleotides.

The numbering of the carbons in these two sugars is worth remembering, as it is the basis for describing the directionality of polynucleotide chains. Primes (') are used to number the sugars because non-primed numbers are used for the bases.

It's also worth noting that ribose is a carbohydrate (with the empirical formula CH_2O), whereas deoxyribose is not.

Each nitrogenous base is a derivative of purine or pyrimidine



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The nitrogenous bases attached to each nucleotide are all derived from one of two simpler compounds: purine or pyrimidine. Purine, whose structure and numbering is shown, is a fused, bicyclic aromatic compound. ('Bicyclic' means the purines are made of two rings. 'Fused' means the 5-atom and 6-atom rings are attached at two contiguous atoms.) In contrast, pyrimidine is a smaller compound, consisting of only a single six-membered ring. Two nitrogenous bases are derived from purine: guanine and adenine. Three nitrogenous bases are derived from pyrimidine: cytosine, thymine, and uracil. In living systems thymine is only found in DNA and uracil is only found in RNA.

All of these bases are aromatic in character and thus flat. You will recall that we observed this property in adenine in Recitation 6. Due to delocalized pi bonding, each non-hydrogen atom in each ring has sp^2 hybridization and thus has trigonal planar electron domain geometry.